certainty that the laws that work here are the same ones that work in far-flung corners of the cosmos. But we do know that should the laws somehow change way out there, it must be *way* out there, because ever more precise astronomical observations have provided ever more convincing evidence that the laws are uniform throughout space, at least the space we can see. This highlights the amazing power of symmetry. We are bound to planet earth and its vicinity. And yet, because of translational symmetry, we can learn about fundamental laws at work in the entire universe without straying from home, since the laws we discover here *are* those laws.

Rotational symmetry or *rotational invariance* is a close cousin of translational invariance. It is based on the idea that every spatial direction is on an equal footing with every other. The view from earth certainly doesn't lead you to this conclusion. When you look up, you see very different things than you do when you look down. But, again, this reflects details of the environment; it is not a characteristic of the underlying laws themselves. If you leave earth and float in deep space, far from any stars, galaxies, or other heavenly bodies, the symmetry becomes evident: there is nothing that distinguishes one particular direction in the black void from another. They are all on a par. You wouldn't have to give a moment's thought to whether a deep-space laboratory you're setting up to investigate properties of matter or forces should be oriented this way or that, since the underlying laws are insensitive to this choice. If one night a prankster were to change the laboratory's gyroscopic settings, causing it to rotate some number of degrees about some particular axis, you'd expect this to have no consequences whatsoever for the laws of physics probed by your experiments. Every measurement ever done fully confirms this expectation. Thus, we believe that the laws that govern the experiments you carry out and explain the results you find are insensitive both to where you are—this is translational symmetry—and to how you happen to be oriented in space—this is rotational symmetry.[1]

As we discussed in Chapter 3, Galileo and others were well aware of another symmetry that the laws of physics should respect. If your deep-space laboratory is moving with constant velocity—regardless of whether you're moving 5 miles per hour this way or 100,000 miles per hour that way—the motion should have absolutely no effect on the laws